

ELK-Em: Closure-Aware Embeddings for $\mathcal{EL}_\perp^+$ Ontologies (Technical Report)

Naman Singh, Ernesto Jiménez-Ruiz and Tillman Weyde

City St George's, University of London, London, UK

Abstract

$\mathcal{EL}_\perp^+$ is a lightweight description logic designed to capture the expressive requirements of large practical ontologies while supporting tractable reasoning, as implemented by reasoners such as ELK. Recent efforts have developed scalable methods for learning continuous representations of such ontologies for inductive inference, but their coverage of the normalised axiom forms varies, they provide no formal guarantees relative to an inference calculus, and they cannot place individuals with no observed assertions. We present ELK-Em, a novel geometric ontology embedding model that embeds (i) concepts as axis-aligned boxes; (ii) roles with a new parametrisation that allows coverage of every normalised $\mathcal{EL}_\perp^+$ axiom form, yielding, at zero loss, a model of the ontology and satisfaction of all of ELK's inference rules, with role chains enforced by a stronger sufficient condition; and (iii) individuals as points regularised using domain-specific encoders. On GALEN, GO, and Anatomy, ELK-Em ranks the atomic subsumptions ELK derives above non-entailed candidates more accurately than geometric baselines. On CAFA5 human protein-function data it also predicts GO concept assertions for held-out proteins from ESM-2 embeddings; nearest-neighbour transfer scores higher, a gap we locate in box-membership scoring rather than placement.

Keywords

Ontology Embeddings, $\mathcal{EL}$ Embeddings, Biomedical Applications

1. Introduction

The $\mathcal{EL}$ family of description logics [1] underlies OWL 2 EL [2], designed for ontologies with large numbers of classes and relations while retaining polynomial-time reasoning for core tasks. This makes it suitable for large biomedical ontologies such as the Gene Ontology (GO) [3], SNOMED CT [4] and GALEN [5]. In these logics, domain knowledge is expressed through axioms, divided into an ABox of instance-level assertions such as $Parent(Alex)$ or $hasChild(Alex, Alice)$, and a TBox of concept-level axioms such as $Parent \sqsubseteq Person$.

The polynomial-time reasoning guarantees enable reasoners such as ELK [6] to perform consistency checking, subsumption and instance checking over large ontologies: the axioms above entail $Person(Alex)$, which ELK derives automatically, letting applications draw on implicit as well as explicit knowledge. Such reasoning is nonetheless limited to the symbolic input. In the absence of ABox assertions for an individual, a reasoner cannot derive its class membership, even if the individual carries labels, comments or external descriptions.

Ontology embedding methods provide one way to support inductive inference over OWL ontologies [7]. Methods such as Onto2Vec [8], OPA2Vec [9] and OWL2Vec* [10] learn vector representations from ontology-derived corpora, but treat ontologies as graph- or corpus-based inputs and do not impose description-logic semantics as constraints during training. More recent works [11, 12, 13, 14, 15, 16] represent concepts, roles and individuals as geometric objects and encode axioms as constraints in the embedding space. These models tie the representation more closely to description-logic reasoning, but three limitations remain:

Third Workshop on Knowledge Graphs and Neurosymbolic AI (KG-NeSy 2026), in ISWC 2026 Workshops Joint Proceedings, October 25–26, 2026, Bari, Italy

✉ naman.singh@citystgeorges.ac.uk (N. Singh); ernesto.jimenez-ruiz@citystgeorges.ac.uk (E. Jiménez-Ruiz); t.e.veyde@citystgeorges.ac.uk (T. Weyde)

ORCID 0009-0000-3659-0266 (N. Singh); 0000-0002-9083-4599 (E. Jiménez-Ruiz); 0000-0001-8028-9905 (T. Weyde)



© 2026 for this paper by its authors. Use permitted under Creative Commons License Attribution 4.0 International (CC BY 4.0).

- They learn individual embeddings only from observed ABox facts, and so cannot place an individual with no assertions, whatever labels, annotations or domain-specific encodings it carries. This limits applications such as CAFA [17], SNOMED CT entity linking [18] and phenotype annotation [19, 20].
- Their geometric designs may impose semantic over-restrictions. As Bourgaux et al. [21] highlight, models might approximate $\exists r.C \sqsubseteq \perp$ as $C \sqsubseteq \perp$, which is not a valid inference and incorrectly penalises every candidate assertion $C(a)$.
- Coverage of the normalised axiom forms varies, with role inclusion and composition treated partially or not at all. Since ABox scores depend on the learned TBox geometry, consequences the geometry does not reflect may lead to poorly scored assertions for individuals.

To address these limitations, we present ELK-Em, a geometric ontology embedding model for $\mathcal{EL}_\perp^+$, the fragment ELK supports, representing concepts as axis-aligned boxes, roles through a parametrisation that allows coverage of all normalised $\mathcal{EL}_\perp^+$ axiom forms, and individuals as points regularised using domain-specific encoders. Boxes are used for concepts because, unlike balls, they are closed under intersection [13]. The role parametrisation yields two guarantees: if every axiom of the ontology has zero geometric loss, then (i) the learned embedding is a model of the ontology, and (ii) it satisfies every consequence derivable by ELK’s inference rules, with role chains enforced by a sufficient condition that we formalise and characterise exactly (Section 4.1). The model is thus a geometric encoding of the logic ELK reasons over, its constructions mirroring ELK’s inference rules. The name reflects that target rather than an implementation dependency, and the reasoner is not invoked during model training.

On GALEN [5], GO [3] and Anatomy (Uberon) [22], ELK-Em ranks the atomic subsumptions ELK derives above non-entailed candidates more accurately than geometric baselines. Using ESM-2 protein-sequence representations [23], it also predicts GO concept assertions for held-out human proteins on CAFA5-derived data [17], whose annotations are withheld during training; we compare against direct sequence-similarity transfer and analyse the results.

The rest of the paper is organised as follows. Section 2 reviews related work on ontology embeddings, with a focus on prior geometric methods for $\mathcal{EL}$. Section 3 introduces the required background on $\mathcal{EL}_\perp^+$. Section 4 presents ELK-Em: its geometric semantics, training, theoretical properties and lexical regularisation. Section 5 evaluates it on ranking entailed subsumptions and on inductive ABox inference, and Section 6 concludes with directions for future work.

2. Related Work

Knowledge graph embedding methods preceded much OWL ontology embedding work and inspired representation learning for structured knowledge [24]. They embed RDF-style multi-relational graphs for link prediction; TransE, for example, embeds a triple (h, r, t) so that $\mathbf{h} + \mathbf{r}$ is close to $\mathbf{t}$ for observed triples and distant for corrupted ones [25]. Later work predicts links for entities unseen during training, typically by encoding textual descriptions [26] or relation-level features [27]. All of these target ABox-level facts, whereas OWL ontologies also contain TBox axioms with class hierarchies, logical constructors, existential restrictions and role inclusions, which requires embedding methods that account for OWL entities, axioms and logical structure.

Ontology embedding methods address this setting [7]. Onto2Vec [8] expands an ontology with inferred axioms using a reasoner, then serialises them into a text-like corpus for Word2Vec training; OPA2Vec [9] augments that corpus with annotation axioms and lexical metadata such as labels, descriptions and synonyms; OWL2Vec* [10] combines random walks over the ontology graph with lexical information and logical constructors. Their objectives do not enforce adherence to the underlying description-logic semantics in the learned representation: axioms and constructors serve as context rather than as constraints. These methods also rely on ontology-internal lexical metadata, which is not always the most informative signal for comparing individuals. In protein-function prediction, amino-acid sequences provide both a domain-specific basis for measuring similarity and a central signal for inferring function, which motivates CAFA’s evaluation of sequence-based annotation methods;

ESM-2 supplies sequence-derived representations for this purpose [17, 23], and we exploit this external signal in Section 4.4.

A second line of work develops geometric models for logics of the $\mathcal{EL}$ family. ELEm uses n -balls for concepts and translations for roles [11], and EmEL⁺⁺ extends this to further constructs [12]. ELBE represents concepts as axis-aligned boxes with translational roles, boxes being closed under intersection [13]; BoxEL also uses boxes but models roles as affine transformations [14]; Box²EL and TransBox further enrich the representation of roles and complex concepts [15, 16], the latter targeting closure under the constructors of the logic. Coverage of the normalised axiom forms varies across these models, with role inclusion, role composition and unsatisfiable existential restrictions treated partially or not at all. They are closer to the model-theoretic structure of the logic than corpus-based embeddings, but establish no guarantees relative to the ELK inference calculus, and individual placement remains tied to observed ABox facts rather than external encoders, leaving individuals with no assertions outside the scope of these models. ELK-Em addresses this gap: geometric modelling of the normalised axiom forms of Section 3 with guarantees relative to ELK inference, and encoder-based placement of individuals with no observed assertions.

3. Background: The Description Logic $\mathcal{EL}_\perp^+$

We consider the tractable description logic $\mathcal{EL}_\perp^+$, a member of the $\mathcal{EL}$ family closely related to the logical core of the OWL 2 EL profile [2] and widely used in large biomedical ontologies such as the Gene Ontology, GALEN, and Uberon. It is the description logic underlying the ELK reasoner [6].

An $\mathcal{EL}_\perp^+$ signature consists of a set N_C of *concept names*, a set N_R of *role names*, and a set N_I of *individual names*. Concept descriptions are built with the constructors in Table 1. An ontology $\mathcal{O} = \mathcal{T} \cup \mathcal{A}$ consists of a TBox $\mathcal{T}$, a finite set of *general concept inclusions* (GCIs) and *role inclusions* (RIs), and an ABox $\mathcal{A}$, a finite set of *concept assertions* $C(a)$ and *role assertions* $r(a, b)$. The members of $\mathcal{T}$ and $\mathcal{A}$ are the *axioms* of $\mathcal{O}$.

The semantics is given by *interpretations*. An interpretation $\mathcal{I} = (\Delta^\mathcal{I}, \cdot^\mathcal{I})$ consists of a non-empty *domain* $\Delta^\mathcal{I}$ and a function $\cdot^\mathcal{I}$ mapping each $a \in N_I$ to an element $a^\mathcal{I} \in \Delta^\mathcal{I}$, each $A \in N_C$ to a set $A^\mathcal{I} \subseteq \Delta^\mathcal{I}$, and each $r \in N_R$ to a relation $r^\mathcal{I} \subseteq \Delta^\mathcal{I} \times \Delta^\mathcal{I}$, extended to complex concepts as in Table 1. $\mathcal{I}$ *satisfies* an axiom α , written $\mathcal{I} \models \alpha$, if the corresponding condition holds, and is a *model* of $\mathcal{O}$, written $\mathcal{I} \models \mathcal{O}$, if it satisfies every axiom of $\mathcal{O}$. $\mathcal{O}$ is *satisfiable* if it has a model. An axiom α is *entailed* by $\mathcal{O}$, written $\mathcal{O} \models \alpha$, if every model of $\mathcal{O}$ satisfies α .

Normal form. A *normalised* $\mathcal{EL}_\perp^+$ ontology contains only GCIs and RIs of the following forms, where

Table 1
Syntax and semantics of the $\mathcal{EL}_\perp^+$ constructors and axioms.

Name	Syntax	Semantics
Top	$\top$	$\Delta^\mathcal{I}$
Bottom	$\perp$	$\emptyset$
Concept name	A	$A^\mathcal{I} \subseteq \Delta^\mathcal{I}$
Conjunction	$C \sqcap D$	$C^\mathcal{I} \cap D^\mathcal{I}$
Existential rest.	$\exists r.C$	$\{x \mid \exists y : (x, y) \in r^\mathcal{I} \wedge y \in C^\mathcal{I}\}$
GCI	$C \sqsubseteq D$	$C^\mathcal{I} \subseteq D^\mathcal{I}$
RI	$r_1 \circ \dots \circ r_k \sqsubseteq s$	$r_1^\mathcal{I} \circ \dots \circ r_k^\mathcal{I} \subseteq s^\mathcal{I}$
Concept assertion	$C(a)$	$a^\mathcal{I} \in C^\mathcal{I}$
Role assertion	$r(a, b)$	$(a^\mathcal{I}, b^\mathcal{I}) \in r^\mathcal{I}$

$A, B, D \in \mathbf{N}_C$ and $r, r_1, r_2, s \in \mathbf{N}_R [1]$:

$$\begin{array}{ll}
\text{GCI0: } A \sqsubseteq B & \text{GCI0-BOT: } A \sqsubseteq \perp \\
\text{GCI1: } A \sqcap B \sqsubseteq D & \text{GCI1-BOT: } A \sqcap B \sqsubseteq \perp \\
\text{GCI2: } A \sqsubseteq \exists r.B & \\
\text{GCI3: } \exists r.A \sqsubseteq B & \text{GCI3-BOT: } \exists r.A \sqsubseteq \perp \\
\text{RI0: } r \sqsubseteq s & \text{RI1: } r_1 \circ r_2 \sqsubseteq s
\end{array}$$

and atomic ABox assertions of the forms $A(a)$ and $r(a, b)$, with role assertions left unchanged. Every $\mathcal{EL}_\perp^+$ ontology can be transformed in linear time into a normalised ontology that is a conservative extension of it [28] (Appendix A.1).

Classification with ELK. ELK [6] classifies an $\mathcal{EL}_\perp^+$ ontology by computing all entailed subsumptions $A \sqsubseteq B$ between named concepts (form GCI0), using the sound and complete inference rules of Figure 1. Section 5.1 evaluates how well an ELK-Em embedding ranks the subsumptions ELK derives above non-entailed candidates, in comparison with geometric embedding baselines.

$$\begin{array}{ll}
R_0 \quad \frac{}{C \sqsubseteq C} & R_\top \quad \frac{}{C \sqsubseteq \top} \\
R_\perp \quad \frac{E \sqsubseteq \exists r.C \quad C \sqsubseteq \perp}{E \sqsubseteq \perp} & R_{\sqcap}^- \quad \frac{C \sqsubseteq D_1 \sqcap D_2}{C \sqsubseteq D_1 \quad C \sqsubseteq D_2} \\
R_{\sqcap}^+ \quad \frac{C \sqsubseteq D_1 \quad C \sqsubseteq D_2}{C \sqsubseteq D_1 \sqcap D_2} & R_{\exists} \quad \frac{E \sqsubseteq \exists r.C \quad C \sqsubseteq D}{E \sqsubseteq \exists r.D} \\
R_\sqsubseteq \quad \frac{C \sqsubseteq D}{C \sqsubseteq E} : D \sqsubseteq E \in \mathcal{O} & R_H \quad \frac{E \sqsubseteq \exists r.C}{E \sqsubseteq \exists s.C} : r \sqsubseteq s \in \mathcal{O} \\
R_\circ \quad \frac{E \sqsubseteq \exists r_1.C \quad C \sqsubseteq \exists r_2.D}{E \sqsubseteq \exists s.D} : r_1 \circ r_2 \sqsubseteq s \in \mathcal{O}
\end{array}$$

Figure 1: Inference rules for classification in $\mathcal{EL}_\perp^+$, following [6]. Expressions above the line are (derived) premises, those below are derived conclusions, and expressions after the colon are side conditions, i.e. fixed axioms of the input ontology whose presence fires the rule.

4. Method

In this section, we introduce the ELK-Em model. Figure 2 depicts an overview of its main architectural components. ELK-Em accepts any OWL ontology as input but preserves only $\mathcal{EL}_\perp^+$ TBox axioms, which are normalised before the geometric embeddings are computed. ABox axioms may be given together with the TBox or in independent files, and the geometric embedding can be combined with a lexical embedding, where available, to improve the placement of unseen instances. We apply ELK-Em to ranking the atomic subsumptions ELK derives and to class-assertion prediction.

The remainder of this section presents the geometric design of ELK-Em, its training procedure, scoring method, formal guarantees, and lexical regularisation strategy. Proofs are collected in Appendix A.2.

4.1. Geometric Interpretation

An *axis-aligned box* in $\mathbb{R}^n$ is a set $[\mathbf{l}, \mathbf{u}] = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{l} \leq \mathbf{x} \leq \mathbf{u}\}$ where $\mathbf{l}, \mathbf{u} \in \mathbb{R}^n$ and the inequalities are taken component-wise. A box is empty if $\exists i$ such that $u_i < l_i$. We write $[-\infty, \infty]$ for $\mathbb{R}^n$ and $[\infty, -\infty]$ for the canonical empty box as notational shorthand. We write $[\mathbf{l}_x, \mathbf{u}_x] \cap [\mathbf{l}_y, \mathbf{u}_y] = [\max(\mathbf{l}_x, \mathbf{l}_y), \min(\mathbf{u}_x, \mathbf{u}_y)]$ for box intersection and $[\mathbf{l}_x, \mathbf{u}_x] \oplus [\mathbf{l}_y, \mathbf{u}_y] = [\mathbf{l}_x + \mathbf{l}_y, \mathbf{u}_x + \mathbf{u}_y]$ to denote box addition.

An ELK-Em embedding is read as an interpretation $\mathcal{E} = (\mathbb{R}^n, \cdot^\mathcal{E})$ assigning

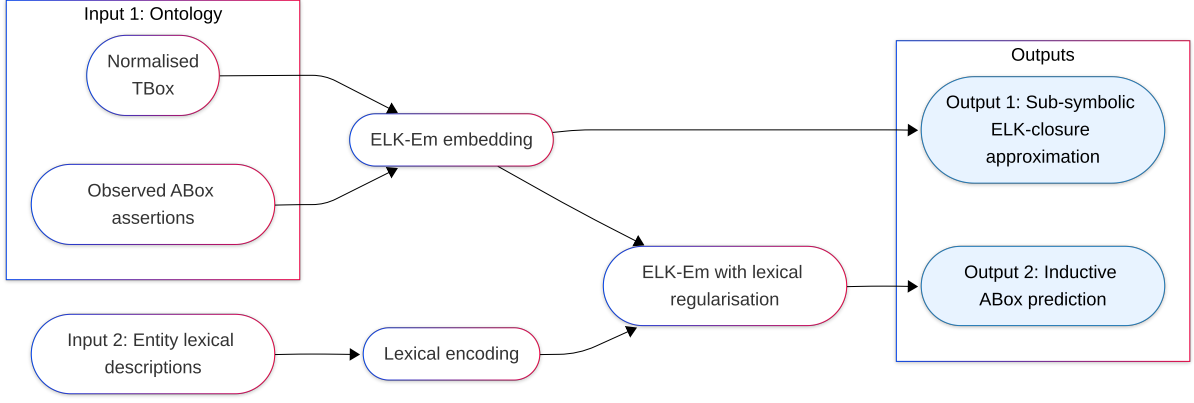


Figure 2: Overview of the ELK-Em pipeline.

- to each $A \in \mathcal{N}_C$ a box $A^\mathcal{E} = B_A = [\mathbf{l}_A, \mathbf{u}_A]$,
- to each $a \in \mathcal{N}_I$ a point $a^\mathcal{E} = \mathbf{e}_a$ (or equivalently a box $[\mathbf{e}_a, \mathbf{e}_a]$),
- to each $r \in \mathcal{N}_R$ a triple $\langle R_r, E_r, T_r \rangle$ with non-empty range box $R_r = [\mathbf{l}_r^R, \mathbf{u}_r^R]$, non-empty error box $E_r = [\mathbf{l}_r^E, \mathbf{u}_r^E]$, and diagonal affine map $T_r(\mathbf{x}) = \mathbf{m}_r \mathbf{x} + \mathbf{c}_r$ ($\mathbf{m}_r \in \mathbb{R}_{>0}^n$, $\mathbf{c}_r \in \mathbb{R}^n$), giving, with $T_r(\mathbf{y})$ read as the singleton box $[T_r(\mathbf{y}), T_r(\mathbf{y})]$, the relation

$$r^\mathcal{E} = \{(\mathbf{x}, \mathbf{y}) \mid \mathbf{x} \in T_r(\mathbf{y}) \oplus E_r \wedge \mathbf{y} \in R_r\}.$$

The interpretation extends to complex concepts following the $\mathcal{EL}_\perp^+$ semantics of Table 1. Top and bottom are the full and empty boxes, conjunction is box intersection, and the existential restriction $\exists r.C$ collects the points related by r to a point lying in both $C^\mathcal{E}$ and the range R_r :

$$\begin{aligned} \top^\mathcal{E} &= \mathbb{R}^n = [-\infty, \infty], & \perp^\mathcal{E} &= \emptyset = [\infty, -\infty], \\ (C \sqcap D)^\mathcal{E} &= C^\mathcal{E} \cap D^\mathcal{E}, & (\exists r.C)^\mathcal{E} &= T_r(C^\mathcal{E} \cap R_r) \oplus E_r. \end{aligned}$$

The last identity holds because $(\exists r.C)^\mathcal{E} = \{\mathbf{x} \mid \exists \mathbf{y} \in C^\mathcal{E}, (\mathbf{x}, \mathbf{y}) \in r^\mathcal{E}\} = \bigcup_{\mathbf{y} \in C^\mathcal{E} \cap R_r} (T_r(\mathbf{y}) \oplus E_r)$, and T_r is a diagonal affine map with positive slope, so it sends boxes to boxes; intersection and $\oplus$ likewise preserve boxes. We write $B_{\exists r.C}$ for this box. The following lemma then follows by induction on the structure of C .

Lemma 1. *For every $\mathcal{EL}_\perp^+$ concept description C , its interpretation $C^\mathcal{E}$ is an axis-aligned box.*

The interpretation determines when the axioms hold in the embedding space (see Figure 3). A concept inclusion $C \sqsubseteq D$ holds when $C^\mathcal{E} \subseteq D^\mathcal{E}$, i.e. $\mathbf{l}_D \leq \mathbf{l}_C \wedge \mathbf{u}_C \leq \mathbf{u}_D$. Similarly, a concept assertion $C(a)$ holds when $a^\mathcal{E} \in C^\mathcal{E}$, i.e. $\mathbf{l}_C \leq \mathbf{e}_a \leq \mathbf{u}_C$. A role assertion $r(a, b)$ holds when $(a^\mathcal{E}, b^\mathcal{E}) \in r^\mathcal{E}$ i.e. $T_r(\mathbf{e}_b) + \mathbf{l}_r^E \leq \mathbf{e}_a \leq T_r(\mathbf{e}_b) + \mathbf{u}_r^E \wedge \mathbf{l}_r^R \leq \mathbf{e}_b \leq \mathbf{u}_r^R$. We now show that each inference rule of Figure 1 has a counterpart at the level of boxes, established in the propositions below and combined in Theorem 8, beginning with existential restriction, which is monotone in its filler C : enlarging C enlarges $\exists r.C$.

Proposition 2. *(Rule $R_{\exists}$.) For any role r and concept descriptions C, D , if $C^\mathcal{E} \subseteq D^\mathcal{E}$, then $(\exists r.C)^\mathcal{E} \subseteq (\exists r.D)^\mathcal{E}$.*

Role inclusions hold when the induced geometric relations are contained: $r \sqsubseteq s$ holds when $r^\mathcal{E} \subseteq s^\mathcal{E}$. The next proposition characterises the containment $r^\mathcal{E} \subseteq s^\mathcal{E}$ through the role parameters.

Proposition 3. *For $r, s \in \mathcal{N}_R$, the following are equivalent:*

1. $r^\mathcal{E} \subseteq s^\mathcal{E}$;
2. $R_r \subseteq R_s$ and $T_r(\mathbf{y}) \oplus E_r \subseteq T_s(\mathbf{y}) \oplus E_s$ for all $\mathbf{y} \in R_r$;
3. $R_r \subseteq R_s$ and $T_r(\mathbf{y}) \oplus E_r \subseteq T_s(\mathbf{y}) \oplus E_s$ for all $\mathbf{y} \in \{\mathbf{l}_r^R, \mathbf{u}_r^R\}$.

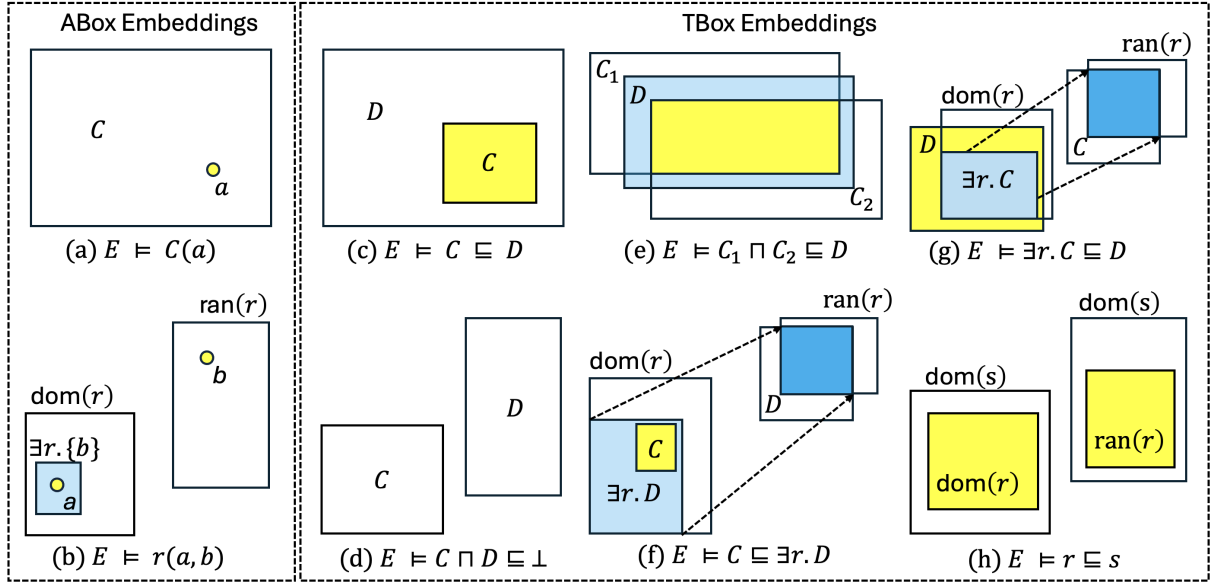


Figure 3: Geometric semantics of ABox axioms (left) and TBox axioms (right) in $\mathcal{EL}_\perp^+$ under ELK-Em embeddings.

The containment also has consequences: it transfers role assertions and existential restrictions from r to s , so that existential restriction is monotone in its role as well as in its filler.

Proposition 4. *If $r^\mathcal{E} \subseteq s^\mathcal{E}$, then*

1. $(a^\mathcal{E}, b^\mathcal{E}) \in r^\mathcal{E}$ implies $(a^\mathcal{E}, b^\mathcal{E}) \in s^\mathcal{E}$;
2. (Rule R_H .) $(\exists r.C)^\mathcal{E} \subseteq (\exists s.C)^\mathcal{E}$ for every concept C .

The treatment of role chain axioms is as follows. For a chain axiom $r_1 \circ r_2 \sqsubseteq s$, satisfaction is containment of the composite in s : $r_1 \circ r_2 \sqsubseteq s$ holds when $r_1^\mathcal{E} \circ r_2^\mathcal{E} \subseteq s^\mathcal{E}$, where

$$r_1^\mathcal{E} \circ r_2^\mathcal{E} = \{(\mathbf{x}, \mathbf{z}) \mid \exists \mathbf{y} : (\mathbf{x}, \mathbf{y}) \in r_1^\mathcal{E} \wedge (\mathbf{y}, \mathbf{z}) \in r_2^\mathcal{E}\}.$$

Proposition 5. *Let $\max(r_1 \circ r_2) = \langle R_{r_2}, \mathbf{m}_{r_1} E_{r_2} \oplus E_{r_1}, T_{r_1} \circ T_{r_2} \rangle^1$ and let $T_{r_2}(R_{r_2}) \oplus E_{r_2} \subseteq R_{r_1}$ be the compatibility condition. Then*

1. $r_1^\mathcal{E} \circ r_2^\mathcal{E} \subseteq \max(r_1 \circ r_2)^\mathcal{E}$;
2. equality holds iff the compatibility condition holds.
3. if the compatibility condition fails, $r_1^\mathcal{E} \circ r_2^\mathcal{E}$ is not of the form $\langle R, E, T \rangle$.

The compatibility condition places the domain $(\exists r_2.\top)^\mathcal{E}$ of r_2 within the range R_{r_1} of r_1 .

Proposition 6. *If $\max(r_1 \circ r_2)^\mathcal{E} \subseteq s^\mathcal{E}$, then*

1. $\mathcal{E} \models r_1 \circ r_2 \sqsubseteq s$;
2. (Rule R_o .) If $X^\mathcal{E} \subseteq (\exists r_1.C)^\mathcal{E}$ and $C^\mathcal{E} \subseteq (\exists r_2.D)^\mathcal{E}$, then $X^\mathcal{E} \subseteq (\exists s.D)^\mathcal{E}$.

The true composite $r_1^\mathcal{E} \circ r_2^\mathcal{E}$ is in general not of the form $\langle R, E, T \rangle$ (Prop. 5.3), so Proposition 3 does not apply to it and containment in s cannot be reduced to a check on parameters. We therefore work with the over-approximation $\max(r_1 \circ r_2)$, which is of that form (Prop. 5.1), and enforce $\max(r_1 \circ r_2)^\mathcal{E} \subseteq s^\mathcal{E}$ in place of the chain axiom. This is sufficient: by Proposition 6 it entails $\mathcal{E} \models r_1 \circ r_2 \sqsubseteq s$ and rule R_o , though it is strictly stronger than the axiom where the over-approximation is strict (Prop. 5.2). Appendix A.3 gives an alternative under-approximation for which R_o fails.

Satisfaction. An interpretation $\mathcal{E}$ satisfies a normalised axiom, written $\mathcal{E} \models \alpha$, according to:

¹We write $\mathbf{m}E$ for the box $[\mathbf{ml}^E, \mathbf{mu}^E]$.

- $\mathcal{E} \models C(a)$ iff $a^\mathcal{E} \in C^\mathcal{E}$
- $\mathcal{E} \models r(a, b)$ iff $(a^\mathcal{E}, b^\mathcal{E}) \in r^\mathcal{E}$
- $\mathcal{E} \models C \sqsubseteq D$ iff $C^\mathcal{E} \subseteq D^\mathcal{E}$
- $\mathcal{E} \models r \sqsubseteq s$ iff $r^\mathcal{E} \subseteq s^\mathcal{E}$
- $\mathcal{E} \models r_1 \circ r_2 \sqsubseteq s$ iff $r_1^\mathcal{E} \circ r_2^\mathcal{E} \subseteq s^\mathcal{E}$

where $C \sqsubseteq D$ ranges over the non- $\perp$ forms GCI0–GCI3 and C, D denote the (possibly complex) concepts occurring in those forms. For chain axioms we enforce the sufficient condition $\max(r_1 \circ r_2)^\mathcal{E} \subseteq s^\mathcal{E}$ in place of the last condition (Prop. 6.1). The bottom cases $A \sqsubseteq \perp$ (GCI0-BOT), $A \sqcap B \sqsubseteq \perp$ (GCI1-BOT), and $\exists r.A \sqsubseteq \perp$ (GCI3-BOT) are semantically covered by the same condition, since $\perp^\mathcal{E} = \emptyset$, but are enforced geometrically through box disjointness rather than inclusion; we treat them via the disjointness losses in Section 4.2.

4.2. Model Training

We train the embedding by minimising a geometric loss whose zero set is contained in the satisfaction conditions above. We first define loss primitives on boxes, then a loss for each normal form, and finally the total objective. All losses are margin-based with a scalar margin $\gamma > 0$, added component-wise where the argument is a vector, and follow previous work in being distance-based [15, 16].

Loss primitives. For boxes $B_x = [\mathbf{l}_x, \mathbf{u}_x]$ and $B_y = [\mathbf{l}_y, \mathbf{u}_y]$, write centres $\mathbf{c} = (\mathbf{u} + \mathbf{l})/2$, offsets $\mathbf{o} = (\mathbf{u} - \mathbf{l})/2$, and the per-dimension separation $d(B_x, B_y) = |\mathbf{c}_x - \mathbf{c}_y| - \mathbf{o}_x - \mathbf{o}_y$, with d_i its i -th component. We use three primitives:

$$\ell_{\subseteq}^\gamma(B_x, B_y) = \frac{1}{n} \|\text{ReLU}(d(B_x, B_y) + 2\mathbf{o}_x + \gamma)\|_1, \quad (1)$$

$$\ell_{\cap+}^\gamma(B_x, B_y) = \frac{1}{n} \|\text{ReLU}(d(B_x, B_y) + \gamma)\|_1, \quad (2)$$

$$\ell_{\cap-}^\gamma(B_x, B_y) = \text{ReLU}(\gamma - \max_i d_i(B_x, B_y)). \quad (3)$$

$\ell_{\subseteq}^\gamma$ penalises violation of $B_x \subseteq B_y$; $\ell_{\cap+}^\gamma$ drives the boxes to overlap; $\ell_{\cap-}^\gamma$ drives them disjoint. We write ℓ^0 for the margin-free version ($\gamma = 0$) of each primitive.

Lemma 7. *Let $\gamma \geq 0$ and let B_x, B_y be boxes. Then (1) $\ell_{\subseteq}^\gamma(B_x, B_y) = 0$ implies $B_x \subseteq B_y$; (2) if B_x, B_y are non-empty, $\ell_{\cap+}^\gamma(B_x, B_y) = 0$ implies $B_x \cap B_y \neq \emptyset$; and (3) if $\gamma > 0$, $\ell_{\cap-}^\gamma(B_x, B_y) = 0$ implies $B_x \cap B_y = \emptyset$. For non-empty B_x, B_y the converses of (1) and (2) hold at $\gamma = 0$, so $\ell_{\subseteq}^0(B_x, B_y) = 0$ iff $B_x \subseteq B_y$ and $\ell_{\cap+}^0(B_x, B_y) = 0$ iff $B_x \cap B_y \neq \emptyset$.*

Per-axiom losses. Each normal form is assigned the loss in Table 2, built so that its margin-free version is zero when the corresponding satisfaction condition holds: box inclusion for concept inclusion, the same primitive applied to the existential image $B_{\exists r.C}$ for existential restrictions, the disjoint primitive for the bottom forms, and the parameter conditions of Proposition 3 for role inclusion and composition – applied in the latter case to $\max(r_1 \circ r_2)$ (Proposition 5). Concepts asserted unsatisfiable ($A \sqsubseteq \perp$, GCI0-BOT) carry no loss: we fix $B_A = [\infty, -\infty]$, the canonical empty box, so that $A^\mathcal{E} = \emptyset$ by construction. For $\exists r.C \sqsubseteq \perp$ (GCI3-BOT) the disjoint primitive enforces $B_C \cap R_r = \emptyset$, hence $(\exists r.C)^\mathcal{E} = \emptyset$, without forcing $B_C = \emptyset$: an unsatisfiable existential restriction does not render C itself unsatisfiable, avoiding the over-restriction in prior box models that approximate this form as $C \sqsubseteq \perp$ [21].

TBox objective. The TBox loss sums the per-axiom losses of Table 2 over all normalised axioms,

$$\mathcal{L}_T = \sum_{\alpha \in \mathcal{T}} \ell(\alpha),$$

where $\ell(\alpha)$ is the loss of the normal form of α . Optimisation is over mini-batches of normalised axioms.

Scoring. At evaluation time we replace each margin loss by its margin-free counterpart. For a per-axiom loss ℓ , let ℓ^0 denote the version obtained by setting $\gamma = 0$ in every primitive occurring in ℓ . For the forms

Table 2

Loss terms for the normalised $\mathcal{EL}_\perp^+$ axiom forms. Overlap terms ($\ell_{\cap+}$) in GCI1 and GCI3 prevent degenerate solutions and keep the constrained boxes non-empty; $\ell_{\text{RIO}}(\max(r_1 \circ r_2), s)$ applies the R10 loss to $\max(r_1 \circ r_2)$ of Prop. 5. Unsatisfiable concepts ($A \sqsubseteq \perp$) are fixed to the empty box rather than trained.

	Axiom	Loss
GCI0	$C \sqsubseteq D$	$\ell_{\sqsubseteq}^\gamma(B_C, B_D)$
(GCI0-BOT)	$C \sqsubseteq \perp$	$-$ (fixed $B_C = [\infty, -\infty]$)
GCI1	$C_1 \sqcap C_2 \sqsubseteq D$	$\ell_{\sqsubseteq}^\gamma(B_{C_1 \cap C_2}, B_D) + \ell_{\cap+}^\gamma(B_{C_1}, B_{C_2})$
(GCI1-BOT)	$C_1 \sqcap C_2 \sqsubseteq \perp$	$\ell_{\cap-}^\gamma(B_{C_1}, B_{C_2})$
GCI2	$C \sqsubseteq \exists r.D$	$\ell_{\sqsubseteq}^\gamma(B_C, B_{\exists r.D})$
GCI3	$\exists r.C \sqsubseteq D$	$\ell_{\sqsubseteq}^\gamma(B_{\exists r.C}, B_D) + \ell_{\cap+}^\gamma(R_r, B_C)$
(GCI3-BOT)	$\exists r.C \sqsubseteq \perp$	$\ell_{\cap-}^\gamma(R_r, B_C)$
R10	$r \sqsubseteq s$	$\ell_{\sqsubseteq}^\gamma(R_r, R_s) + \sum_{y \in \{1_r^R, u_r^R\}} \ell_{\sqsubseteq}^\gamma(T_r(y) \oplus E_r, T_s(y) \oplus E_s)$
R11	$r_1 \circ r_2 \sqsubseteq s$	$\ell_{\text{RIO}}(\max(r_1 \circ r_2), s)$

whose loss is built from $\ell_{\sqsubseteq}$ and $\ell_{\cap+}$, Lemma 7 gives $\ell^0(\alpha) = 0$ iff the enforced geometric condition for α holds. We score an axiom by

$$s(\alpha) = \exp(-\ell^0(\alpha)) \in (0, 1], \quad (4)$$

so that $s(\alpha) = 1$ when that condition holds and decreases smoothly with the degree of violation. This score is the quantity ranked in the subsumption-ranking evaluation of Section 5.1.

Negative sampling. To avoid degenerate solutions we contrast each positive axiom with corrupted ones. For a positive GCI0 axiom $C \sqsubseteq D$ we generate $k_{\text{neg}} \geq 1$ negatives by replacing D with a uniformly sampled concept name, without filtering corruptions that happen to be entailed; for a positive GCI2 axiom $C \sqsubseteq \exists r.D$ we corrupt C or D with equal probability, keeping r fixed. For each positive or corrupted axiom α we compute its margin-free score $s(\alpha) = \exp(-\ell^0(\alpha))$ from Eq. (4). Let $\sigma \in \mathbb{R}^N$ collect these scores and $\mathbf{t} \in \{0, 1\}^N$ the corresponding binary labels, with sample means $\bar{\sigma}, \bar{t}$. We use a Pearson correlation loss

$$\mathcal{L}_{\text{contrast}} = 1 - \frac{(\sigma - \bar{\sigma})^\top (\mathbf{t} - \bar{t})}{\|\sigma - \bar{\sigma}\| \|\mathbf{t} - \bar{t}\| + \varepsilon}, \quad (5)$$

where $\varepsilon > 0$ prevents division by zero. Minimising $\mathcal{L}_{\text{contrast}}$ encourages separation between positive and corrupted axioms, assigning higher scores to positives and lower scores to corruptions. This provides a lenient contrastive signal without an additional margin hyperparameter while adhering to $\mathcal{EL}_\perp^+$ semantics. It is used only when training on TBox data alone (Section 5.1), where axiom scores are the ranked quantity; with ABox supervision the assertion losses below already supply a contrastive signal.

ABox training. Concept assertions are trained as a special case of GCI0: an individual is embedded as the degenerate box $[\mathbf{e}_a, \mathbf{e}_a]$, which reduces membership $\mathbf{e}_a \in B_C$ to box inclusion. Since ontologies typically provide only positive assertions, we generate negatives by corrupting the class of a positive assertion, filtering out observed training positives. Positive and corrupted assertions are trained with

$$\ell_{C(a)} = \ell_{\sqsubseteq}^\gamma([\mathbf{e}_a, \mathbf{e}_a], B_C), \quad \ell_{\neg C(a)} = \text{ReLU}(\delta - \ell_{\sqsubseteq}^\gamma([\mathbf{e}_a, \mathbf{e}_a], B_C)),$$

where $\delta = -\log \tau$ for a target-score threshold $\tau \in (0, 1)$, giving

$$\mathcal{L}_A = \sum_{C(a) \in \mathcal{A}^+} \ell_{C(a)} + \sum_{\neg C(a) \in \mathcal{A}^-} \ell_{\neg C(a)}, \quad (6)$$

over the sets $\mathcal{A}^+$ and $\mathcal{A}^-$ of positive and corrupted assertions. Role assertions admit the same treatment: the condition $\mathcal{E} \models r(a, b)$ of Section 4.1 is an inclusion between $[\mathbf{e}_a, \mathbf{e}_a]$ and $T_r([\mathbf{e}_b, \mathbf{e}_b]) \oplus E_r$ together with $\mathbf{e}_b \in R_r$, both expressible with $\ell_{\sqsubseteq}^\gamma$. We restrict supervision to concept assertions here, matching

our inductive task of predicting concept membership for individuals without assertions (Section 5.2); role-assertion supervision, such as training on protein–protein interaction data alongside functional annotations, is left to future work.

4.3. Theoretical Properties

We now state the main formal properties of ELK-Em. These results show that zero geometric loss is not merely an optimisation criterion, but entails semantic satisfaction of the embedded ontology. Bourgaux et al. [21] show that, among the geometric embedding methods they survey, BoxEL is the only one with a soundness guarantee [14], and it does not cover role inclusion or composition; to our knowledge no prior method relates the learned embedding to closure under ELK’s inference rules. ELK-Em provides both, for all normalised forms and all rules of Figure 1 (proofs in Appendix A.4, complexity in Appendix A.5).

Theorem 8. *Let $\mathcal{O}$ be a normalised ontology, and let $\mathcal{E}$ be an embedding such that every TBox axiom and every positive ABox assertion in $\mathcal{O}$ has zero geometric loss. Then $\mathcal{E}$ is a model of $\mathcal{O}$.*

Corollary 9. *Under the assumptions of Theorem 8, $\mathcal{E} \models \alpha$ for every axiom α derivable from $\mathcal{O}$ by the rules of Figure 1.*

The question of completeness is open: the RI1 condition is sufficient but not necessary, so not every consistent ontology need admit a zero-loss embedding.

4.4. Lexical Regularisation

Many prediction settings involve individuals that lack ABox assertions but carry informative non-logical descriptions, and in which those descriptions are the primary signal for prediction. CAFA predicts GO terms for proteins from their sequence data [17]; the SNOMED CT Entity Linking Challenge assigns SNOMED CT concepts to clinical discharge summaries [18]; rare-disease resources such as CADA and PhEval support phenotype-driven prediction from clinical descriptions [19, 20]; and MONDO associates diseases with names, synonyms, definitions and textual metadata [29]. Across these settings, external descriptions carry information that can position a new individual relative to a curated TBox.

We use lexical regularisation to transfer this information into the ontology embedding space; we take *lexical* to cover any textual description of an individual, such as an amino acid sequence or a clinical narrative, rather than only the ontology labels and annotations the term often denotes. Let $a_1, \dots, a_m$ be a batch of individuals. For each a_i , let $\mathbf{z}_i \in \mathbb{R}^d$ denote an external representation obtained from information available at prediction time, and let $\mathbf{e}_i \in \mathbb{R}^n$ denote its learned point embedding. The external representation may be produced by a task-specific encoder, such as ESM-2 for protein sequences [23] or a Sentence-BERT-style encoder for clinical text [30]. External similarity is computed by cosine similarity,

$$s_{ij}^{\text{ext}} = \frac{\mathbf{z}_i^\top \mathbf{z}_j}{\|\mathbf{z}_i\|_2 \|\mathbf{z}_j\|_2} \in [-1, 1], \quad (7)$$

which is insensitive to the magnitudes of the representations, and rescaled to $[0, 1]$ as $\bar{s}_{ij}^{\text{ext}} = \frac{1}{2}(1 + s_{ij}^{\text{ext}})$. Geometric similarity is defined from the squared Euclidean distance between the embeddings of individuals:

$$s_{ij}^{\text{geo}} = \exp\left(-\frac{\lambda}{n}\|\mathbf{e}_i - \mathbf{e}_j\|_2^2\right), \quad \lambda > 0, \quad (8)$$

which is high for nearby embeddings and decays smoothly with distance; n normalises by the embedding dimension and λ controls the decay rate. The lexical regularisation loss aligns the pairwise similarity structure of the external descriptions with the geometry of the learned embeddings:

$$\mathcal{L}_{\text{lex}} = \frac{1}{\binom{m}{2}} \sum_{1 \leq i < j \leq m} (\bar{s}_{ij}^{\text{ext}} - s_{ij}^{\text{geo}})^2. \quad (9)$$

This encourages individuals with similar external descriptions to be placed close together in the embedding space. For individuals with ABox assertions the term acts as an auxiliary regulariser; for individuals with no observed assertions it provides the sole placement signal, positioning them by external similarity to annotated individuals. The lexical term is the only pairwise cost: for a batch of m individuals it evaluates $\binom{m}{2}$ similarities, each $O(n)$, giving $O(m^2n)$ per batch, independent of ontology size. In Section 5.2, we evaluate this strategy for protein function prediction using ESM-2 sequence representations [23].

5. Evaluation

We first assess how well ELK-Em ranks the entailed atomic subsumptions derived by ELK, and then evaluate it on inductive protein function prediction using protein-sequence encodings as the lexical regularisation input.

5.1. Ranking Entailed Atomic Subsumptions

We evaluate how well the axiom scores of Eq. (4) separate the entailed atomic subsumptions $A \sqsubseteq B$ (form GCI0) derived by ELK from non-entailed candidates, following the setup of Box²EL [15]. The task measures *entailment fidelity*: how far the geometry aligns with the entailment relation, not whether it reproduces classification, which a reasoner already performs exactly.

Datasets and baselines. We use the benchmark and evaluation protocol of Box²EL [15], over the biomedical ontologies GALEN, GO, and Anatomy (Uberon). Models are trained on the normalised ontology used for this task, whose class and role signatures are 23,141/950 for GALEN, 45,893/8 for GO, and 106,258/166 for Anatomy; Table 6 reports axiom counts by normal form. Entailed atomic subsumptions are computed with the ELK reasoner [6]. After removing those already asserted in the training axioms, the benchmark retains 18,138 for GALEN, 3,309 for GO, and 2,130 for Anatomy, split 90/10 into test and validation. Baseline results for ELEM [11], EmEL⁺⁺ [12], BoxEL [14], ELBE [13] and Box²EL are reproduced from Table 6 of [15], who retrain all five models under this protocol rather than adopting previously published numbers. TransBox [16] does not report results under this protocol and is not included.

Evaluation protocol. For each held-out entailed subsumption $C \sqsubseteq D$, the query concept C is fixed and the entire concept signature is ranked in descending order of subsumption score as candidate superclasses. ELK-Em ranks by the margin-free score $s(\alpha) = \exp(-\ell^0(\alpha))$ of Eq. (4); the baselines rank by their own GCI0 scoring functions, based on the distance between concept-box centres [15]. We report Mean Rank (MR), Mean Reciprocal Rank (MRR), Median Rank (Med), AUC and Hits@ k for $k \in \{1, 10, 100\}$ under the unfiltered protocol of [15], retained for comparability with the published baselines.

Experimental protocol. ELK-Em uses the negative sampling and Pearson contrastive objective of Section 4.2, with a single configuration shared across all three ontologies: $n = 50$, Adam, learning rate 0.01 decayed by 0.1 after 5,000 epochs, margin $\gamma = 0.1$, batch size 1,024, 10,000 epochs. Only the number of corrupted axioms per positive differs ($k_{\text{neg}} = 5$ for GO, $k_{\text{neg}} = 3$ for GALEN and Anatomy). Checkpoints are selected by median rank on the full validation split, evaluated every 100 epochs. Training takes 6–7 minutes per run on a single A100 with 1.7 GB peak host memory, giving models of 2.6M, 4.6M and 10.7M parameters for GALEN, GO and Anatomy. Results are means over five random seeds.

Results. Two features of the benchmark shape Table 3. The unfiltered protocol leaves alternative asserted or entailed superclasses in the candidate list, so the ranking space mixes false negatives, genuine non-entailments and consistent non-consequences. A single competing superclass displaces the target from the top of the ranking, which is why Hits@1 is low for all models. Role compositions are also sparse (58, 6 and 31 axioms), so the results say little about RI1 in isolation, though a single composition licenses $R_{\circ}$ derivations across every pair of existential restrictions it connects. With that in mind, ELK-Em

Table 3

Ranking of entailed atomic subsumptions on GALEN, GO, and Anatomy. Baseline results are from Table 6 of [15] and use $n = 200$ ($n = 25$ for BoxEL) with per-dataset hyperparameter tuning; ELK-Em uses $n = 50$ and a single configuration across all three ontologies. Both ELK-Em and the baselines are means over five random seeds; deviations are shown for ELK-Em only, as [15] reports means alone. Deviations below 0.005 display as 0.00. Best value per column and ontology in bold, with ties at the displayed precision bolded jointly.

	Model	H@1 $\uparrow$	H@10 $\uparrow$	H@100 $\uparrow$	Med $\downarrow$	MRR $\uparrow$	MR $\downarrow$	AUC $\uparrow$
GALEN	ELEm	0.00	0.04	0.20	1807	0.01	4405	0.81
	EmEL ⁺⁺	0.00	0.04	0.18	2049	0.01	4634	0.81
	BoxEL	0.00	0.00	0.01	6906	0.00	7925	0.67
	ELBE	0.00	0.06	0.16	1785	0.02	3974	0.84
	Box ² EL	0.01	0.09	0.24	1003	0.03	2833	0.88
	ELK-Em	0.00 $\pm$ 0.00	0.08 $\pm$ 0.00	0.35 $\pm$ 0.00	226 $\pm$ 3	0.03 $\pm$ 0.00	2323 $\pm$ 30	0.90 $\pm$ 0.00
GO	ELEm	0.00	0.04	0.22	1629	0.02	7377	0.84
	EmEL ⁺⁺	0.00	0.04	0.19	1346	0.01	6557	0.86
	BoxEL	0.00	0.00	0.13	1085	0.00	5359	0.88
	ELBE	0.00	0.06	0.21	935	0.02	3846	0.92
	Box ² EL	0.00	0.08	0.49	107	0.04	1689	0.96
	ELK-Em	0.00 $\pm$ 0.00	0.10 $\pm$ 0.01	0.71 $\pm$ 0.01	43 $\pm$ 1	0.04 $\pm$ 0.00	529 $\pm$ 51	0.99 $\pm$ 0.00
Anatomy	ELEm	0.00	0.07	0.28	901	0.02	7958	0.93
	EmEL ⁺⁺	0.00	0.07	0.26	1576	0.02	10976	0.90
	BoxEL	0.01	0.10	0.24	838	0.04	9156	0.92
	ELBE	0.00	0.08	0.32	336	0.03	2312	0.98
	Box ² EL	0.01	0.09	0.44	152	0.04	1599	0.99
	ELK-Em	0.03 $\pm$ 0.00	0.20 $\pm$ 0.01	0.50 $\pm$ 0.01	104 $\pm$ 3	0.08 $\pm$ 0.00	1368 $\pm$ 95	0.99 $\pm$ 0.00

improves on the geometric baselines across all three ontologies, most consistently in median and mean rank, the only exceptions being Hits@1 and Hits@10 on GALEN, using a single configuration at $n = 50$ against baselines tuned per dataset at $n = 200$, with seed variance small relative to the gaps.

5.2. Inductive Protein Function Prediction

We report preliminary results on inductive ABox prediction: assigning GO concepts to proteins that contribute no assertions during training. The aim is to test whether such individuals can be placed at all, not to offer competitive protein function prediction results. This is the setting that motivates lexical regularisation, and it lies outside the reach of a reasoner, which derives nothing for an individual with no assertions. Prior geometric embedding models learn individual embeddings from observed ABox facts alone, so to our knowledge this setting has not been evaluated for them.

Dataset and baseline. We use a CAFA5-derived benchmark over human proteins and Gene Ontology annotations [17], comprising 25,125 proteins split by protein identity into 20,100 training, 2,512 validation and 2,513 test proteins. Validation and test proteins are held out at the individual level, so no protein-function assertion for them appears in the training ABox. Their ESM-2 representations are available during training, so the setting is inductive with respect to assertions rather than to individuals. The candidate set comprises the 43,248 GO concepts of the ontology,² with no filtering by sub-ontology or annotation frequency. External representations are 480-dimensional ESM-2 (t12, 35M) embeddings, mean-pooled over residues (Appendix A.7). Sequence-similarity transfer has been a reference method in CAFA evaluations since the first challenge and remains difficult to improve on [31]; we compare against transferring annotations from the k_{nn} nearest training proteins in ESM-2 space, with no ontology embedding involved. Both methods consume the same ESM-2 vectors, annotations, candidate set and threshold selection, so the comparison isolates the scoring mechanism.

Evaluation protocol. We follow the protein-centric CAFA protocol [31], detailed in Appendix A.8. Predictions and ground truth are both propagated through the GO hierarchy before scoring. Thresholds

²Fewer than $|N_C| = 45,893$ in Table 6: fresh names from normalisation are embedded but are not candidates.

Table 4Inductive protein-function prediction, means over three seeds ($\pm$ standard deviation).

Setting	Precision	Recall	F_1
ELK-Em w/o lexical regularisation	0.0616 $\pm$ 0.0010	0.8371 $\pm$ 0.0044	0.1148 $\pm$ 0.0017
ELK-Em	0.2208 $\pm$ 0.0143	0.6135 $\pm$ 0.0083	0.3246 $\pm$ 0.0165
ESM-2 nearest-neighbour transfer	0.6083	0.5788	0.5932

are swept over 71 values evenly spaced in $[0.3, 1.0]$; the threshold maximising validation F_1 is applied unchanged to the test split, where we report precision, recall and F_1 . For the baseline, both k_{nn} and the threshold are selected on validation alone.

Experimental protocol. ELK-Em is trained in dimension $n = 100$ for 15,000 epochs with Adam at learning rate 0.01, held constant for the first third of training and then cosine-decayed to 10^{-4} . TBox and ABox axioms are batched at 1,024, with margin $\gamma = 0.05$, two corrupted assertions per positive and target-score threshold $\tau = 0.4$. Lexical regularisation uses batches of 2,000 individuals with decay rate $\lambda = 1.0$. The ABox positive, negative and lexical terms are weighted 2.0, 0.5 and 10.0 respectively, the last selected on validation from $\{1, 10, 50, 200\}$. Results are means over three random seeds ($\pm$ standard deviation).

Results. Table 4 reports the effect of lexical regularisation. Without it, held-out proteins have no evidence tying them to the training proteins: their embeddings remain close to initialisation and many are assigned broad, high-level GO concepts, giving high recall at very low precision. With it, proteins are positioned by sequence similarity to annotated training proteins and F_1 improves roughly threefold, so inductive placement is possible for individuals with no observed assertions. The baseline nonetheless reaches $F_1 = 0.59$ and outperforms ELK-Em here.

The gap is not attributable to the weight on $\mathcal{L}_{lex}$, which peaks at the value used above (Appendix A.9). Nor does $\mathcal{L}_{lex}$ fail to transfer external structure, though it transfers it unevenly: rank correlation between $\bar{s}^{ext}$ and s^{geo} is 0.96 between test and training individuals, while top- k_{nn} neighbourhood overlap is only 10–27%. Global ordering is preserved, local structure is not, which accounts for part of the difference between nearest-neighbour transfer in ESM-2 space ($F_1 = 0.59$) and in the learned space (0.43). The remainder lies in the score: box-membership scoring on the same embeddings reaches 0.32. It saturates, assigning 1 to every satisfied assertion, and is non-competitive, each of the 4.3×10^4 candidates being tested against its own box in isolation, so an individual may score 1 on arbitrarily many concepts where neighbour transfer distributes fixed weight among them. Both follow from membership being a predicate: what makes the embedding a model is also what makes its scores poor for ranking a large candidate set.

6. Conclusions and Future Work

We presented ELK-Em, a geometric embedding model for $\mathcal{EL}_{\perp}^+$ ontologies. Its parametrisation of roles allows coverage of all normalised axiom forms and yields two guarantees at zero loss: the embedding is a model of the ontology, and it satisfies every consequence derivable by ELK’s inference rules, with role chains enforced through a sufficient condition on an over-approximation of the composite. On GALEN, GO, and Anatomy, ELK-Em ranks the atomic subsumptions ELK derives above non-entailed candidates more accurately than geometric baselines. ELK-Em also places individuals that contribute no assertions during training. On CAFA5-derived human protein-function data, lexical regularisation with ESM-2 representations improves F_1 roughly threefold, though direct sequence-similarity transfer still outperforms it; the difference lies in the scoring rather than the placement, since box membership is a predicate and so ranks a large candidate set poorly.

In the near future, we plan to (i) develop a graded score compatible with the model-theoretic semantics, (ii) study whether closer capture of $\mathcal{EL}_{\perp}^+$ semantics improves general TBox prediction, beyond the

entailed subsumptions evaluated here, (iii) extend lexical regularisation to the inductive placement of concepts and roles with unseen TBox axioms, (iv) analyse more expressive role representations admitting tighter treatments of composition, (v) extend ELK-Em to tasks such as protein–protein interaction prediction, and (vi) design metrics that assess deductive-closure approximation more accurately, such as precision and recall against the ELK closure in place of ranking under an unfiltered protocol.

Acknowledgments

Ernesto was partially supported by *GUARD*. AI tools were used only for grammar and minor language edits, not for research design, experiments, analysis, or the development of ideas.

References

- [1] F. Baader, S. Brandt, C. Lutz, Pushing the EL envelope, in: *Proceedings of the 19th International Joint Conference on Artificial Intelligence, IJCAI’05*, Morgan Kaufmann Publishers Inc., San Francisco, CA, USA, 2005, p. 364–369.
- [2] B. Motik, B. Cuenca Grau, I. Horrocks, Z. Wu, A. Fokoue, C. Lutz, *OWL 2 Web Ontology Language Profiles*, W3C Recommendation, World Wide Web Consortium, 2012. URL: <https://www.w3.org/TR/owl2-profiles/>.
- [3] S. Aleksander, J. Balhoff, S. Carbon, et al., The Gene Ontology knowledgebase in 2026, *Nucleic Acids Research* 54 (2025). doi:10.1093/nar/gkaf1292.
- [4] International Health Terminology Standards Development Organisation - IHTSDO, SNOMED CT, 2014. URL: <http://www.ihtsdo.org/snomed-ct/>.
- [5] A. Rector, J. Rogers, Ontological and Practical Issues in Using a Description Logic to Represent Medical Concept Systems: Experience from GALEN, in: *Reasoning Web*, 2006, pp. 197–231. doi:10.1007/11837787_9.
- [6] Y. Kazakov, M. Krötzsch, F. Simančík, The Incredible ELK, *J. Autom. Reason.* 53 (2014) 1–61. doi:10.1007/s10817-013-9296-3.
- [7] J. Chen, O. Mashkova, F. Zhapa-Camacho, R. Hoehndorf, Y. He, I. Horrocks, Ontology Embedding: A Survey of Methods, Applications and Resources, *IEEE Transactions on Knowledge and Data Engineering* 37 (2025) 4193–4212. doi:10.1109/tkde.2025.3559023.
- [8] F. Z. Smaili, X. Gao, R. Hoehndorf, Onto2Vec: joint vector-based representation of biological entities and their ontology-based annotations, *Bioinformatics* 34 (2018) i52–i60. doi:10.1093/bioinformatics/bty259.
- [9] F. Z. Smaili, X. Gao, R. Hoehndorf, OPA2Vec: combining formal and informal content of biomedical ontologies to improve similarity-based prediction, *Bioinformatics* 35 (2019) 2133–2140. doi:10.1093/bioinformatics/bty933.
- [10] J. Chen, P. Hu, E. Jimenez-Ruiz, O. M. Holter, D. Antonyrajah, I. Horrocks, OWL2Vec*: Embedding of OWL Ontologies, *Machine Learning* 110 (2021) 1813–1845. doi:10.1007/s10994-021-05997-6.
- [11] M. Kulmanov, W. Liu-Wei, Y. Yan, R. Hoehndorf, EL Embeddings: Geometric Construction of Models for the Description Logic EL++, in: *Proceedings of the Twenty-Eighth International Joint Conference on Artificial Intelligence, IJCAI-19*, International Joint Conferences on Artificial Intelligence Organization, 2019, pp. 6103–6109. doi:10.24963/ijcai.2019/845.
- [12] S. Mondal, S. K. Bhatia, R. Mutharaju, EmEL++: Embeddings for EL++ Description Logic, in: *AAAI Spring Symposium on Combining Machine Learning with Knowledge Engineering*, 2021.
- [13] X. Peng, Z. Tang, M. Kulmanov, K. Niu, R. Hoehndorf, Description Logic EL++ Embeddings with Intersectional Closure, 2022. URL: <https://arxiv.org/abs/2202.14018>. arXiv:2202.14018.
- [14] B. Xiong, N. Potyka, T.-K. Tran, M. Nayyeri, S. Staab, Faithful Embeddings for EL++ Knowledge Bases, in: *The Semantic Web – ISWC 2022*, volume 13489 of *Lecture Notes in Computer Science*, Springer, Cham, 2022, pp. 22–38. doi:10.1007/978-3-031-19433-7_2.

- [15] M. Jackermeier, J. Chen, I. Horrocks, Dual Box Embeddings for the Description Logic EL++, in: *Proceedings of the ACM Web Conference 2024 (WWW '24)*, Association for Computing Machinery, New York, NY, USA, 2024, pp. 2250–2258. doi:10.1145/3589334.3645648.
- [16] H. Yang, J. Chen, U. Sattler, TransBox: EL++-closed Ontology Embedding, in: *Proceedings of the ACM Web Conference 2025 (WWW '25)*, Association for Computing Machinery, New York, NY, USA, 2025, pp. 22–34. doi:10.1145/3696410.3714672.
- [17] I. Friedberg, P. Radivojac, C. D. Paolis, D. Piovesan, P. Joshi, W. Reade, A. Howard, CAFA 5 Protein Function Prediction, <https://kaggle.com/competitions/cafa-5-protein-function-prediction>, 2023. Kaggle.
- [18] R. Davidson, W. Hardman, G. Amit, et al., SNOMED CT entity linking challenge, *Journal of the American Medical Informatics Association* 32 (2025) 1397–1406. doi:10.1093/jamia/ocaf104.
- [19] C. Peng, S. Dieck, A. Schmid, et al., CADA: phenotype-driven gene prioritization based on a case-enriched knowledge graph, *NAR Genomics and Bioinformatics* 3 (2021) lqab078. doi:10.1093/nargab/lqab078.
- [20] Y. Bridges, V. Souza, K. Cortes, et al., Towards a standard benchmark for phenotype-driven variant and gene prioritisation algorithms: PhEval - Phenotypic inference Evaluation framework, *BMC Bioinformatics* 26 (2025). doi:10.1186/s12859-025-06105-4.
- [21] C. Bourgaux, R. Guimarães, R. Koudijs, V. Lacerda, A. Ozaki, Knowledge Base Embeddings: Semantics and Theoretical Properties, in: *Proceedings of the 21st International Conference on Principles of Knowledge Representation and Reasoning*, 2024, pp. 823–833. doi:10.24963/kr.2024/77.
- [22] C. J. Mungall, C. Torniai, G. V. Gkoutos, S. E. Lewis, M. A. Haendel, Uberon, an integrative multi-species anatomy ontology, *Genome Biology* 13 (2012) R5. doi:10.1186/gb-2012-13-1-r5.
- [23] Z. Lin, H. Akin, R. Rao, et al., Evolutionary-scale prediction of atomic-level protein structure with a language model, *Science* 379 (2023) 1123–1130. doi:10.1126/science.ade2574.
- [24] Q. Wang, Z. Mao, B. Wang, L. Guo, Knowledge Graph Embedding: A Survey of Approaches and Applications, *IEEE Transactions on Knowledge and Data Engineering* 29 (2017) 2724–2743. doi:10.1109/TKDE.2017.2754499.
- [25] A. Bordes, N. Usunier, A. Garcia-Durán, J. Weston, O. Yakhnenko, Translating embeddings for modeling multi-relational data, in: *Proceedings of the 27th International Conference on Neural Information Processing Systems - Volume 2, NIPS'13*, Curran Associates Inc., Red Hook, NY, USA, 2013, p. 2787–2795.
- [26] D. Daza, M. Cochez, P. Groth, Inductive Entity Representations from Text via Link Prediction, in: *Proceedings of the Web Conference 2021 (WWW '21)*, Association for Computing Machinery, New York, NY, USA, 2021, pp. 798–808. doi:10.1145/3442381.3450141.
- [27] G. A. Gesese, H. Sack, M. Alam, RAILD: Towards Leveraging Relation Features for Inductive Link Prediction In Knowledge Graphs, in: *Proceedings of the 11th International Joint Conference on Knowledge Graphs (IJCKG '22)*, Association for Computing Machinery, New York, NY, USA, 2022, pp. 82–90. doi:10.1145/3579051.3579066.
- [28] C. Lutz, D. Walther, F. Wolter, Conservative extensions in expressive description logics, in: *Proceedings of the 20th International Joint Conference on Artificial Intelligence, IJCAI'07*, Morgan Kaufmann Publishers Inc., San Francisco, CA, USA, 2007, p. 453–458.
- [29] N. A. Vasilevsky, S. Toro, N. Matentzoglou, J. E. Flack, K. R. Mullen, H. Hegde, S. Gehrke, P. L. Whetzel, Y. Shwetar, N. L. Harris, et al., Mondo: integrating disease terminology across communities, *Genetics* (2025) iyaf215. doi:10.1093/genetics/iyaf215.
- [30] N. Reimers, I. Gurevych, Sentence-BERT: Sentence embeddings using Siamese BERT-networks, in: *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, Association for Computational Linguistics, Hong Kong, China, 2019, pp. 3982–3992. doi:10.18653/v1/D19-1410.
- [31] Y. Jiang, T. R. Oron, W. T. Clark, et al., An expanded evaluation of protein function prediction methods shows an improvement in accuracy, *Genome Biology* 17 (2016). doi:10.1186/

A. Appendix

A.1. Normalisation Rules

Table 5 lists the normalisation rules for $\mathcal{EL}_\perp^+$ [1]. Exhaustive application terminates in linear time and yields an ontology containing only the forms GCI0–GCI3, GCI0-BOT, GCI1-BOT, GCI3-BOT, RI0 and RI1 of Section 3, together with atomic assertions $A(a)$ and $r(a, b)$. Fresh concepts and roles are embedded exactly as symbols of the input signature.

Table 5

Normalisation rules for $\mathcal{EL}_\perp^+$. C, D, E range over concept descriptions, $\hat{C}, \hat{D}$ over complex (non-atomic) concept descriptions, $r, r_1, \dots, r_k, s$ over role names; A is a fresh concept name and u a fresh role name.

	Input		Output
N1	$r_1 \circ \dots \circ r_k \sqsubseteq s, k > 2$	$\rightsquigarrow$	$r_1 \circ \dots \circ r_{k-1} \sqsubseteq u, u \circ r_k \sqsubseteq s$
N2	$C \sqcap \hat{D} \sqsubseteq E$	$\rightsquigarrow$	$\hat{D} \sqsubseteq A, C \sqcap A \sqsubseteq E$
N3	$\exists r.\hat{C} \sqsubseteq D$	$\rightsquigarrow$	$\hat{C} \sqsubseteq A, \exists r.A \sqsubseteq D$
N4	$\perp \sqsubseteq D$	$\rightsquigarrow$	discard
N5	$\hat{C} \sqsubseteq \hat{D}$	$\rightsquigarrow$	$\hat{C} \sqsubseteq A, A \sqsubseteq \hat{D}$
N6	$D \sqsubseteq \exists r.\hat{C}$	$\rightsquigarrow$	$D \sqsubseteq \exists r.A, A \sqsubseteq \hat{C}$
N7	$E \sqsubseteq C \sqcap D$	$\rightsquigarrow$	$E \sqsubseteq C, E \sqsubseteq D$
N8	$C(a), C \notin \mathbf{N}_C$	$\rightsquigarrow$	$A(a), A \sqsubseteq C$

The normalised ontology $\mathcal{O}'$ is linear in the size of the input $\mathcal{O}$, and conservatively extends it: every model of $\mathcal{O}'$, restricted to the signature of $\mathcal{O}$, is a model of $\mathcal{O}$, and every model of $\mathcal{O}$ can be expanded to a model of $\mathcal{O}'$ [28].

A.2. Proofs for Sections 4.1 and 4.2

Lemma 1. For every $\mathcal{EL}_\perp^+$ concept description C , its interpretation $C^\mathcal{E}$ is an axis-aligned box.

Proof. By induction on C . The base cases $A^\mathcal{E} = B_A$, $\top^\mathcal{E}$ and $\perp^\mathcal{E}$ are boxes by definition, and box intersection is a box, giving $C \sqcap D$. For $\exists r.C$, the box $C^\mathcal{E} \cap R_r$ is mapped to a box by T_r , which is diagonal affine with positive slope, and $\oplus E_r$ preserves boxes. $\square$

Proposition 2. For any role r and concept descriptions C, D , if $C^\mathcal{E} \subseteq D^\mathcal{E}$, then $(\exists r.C)^\mathcal{E} \subseteq (\exists r.D)^\mathcal{E}$.

Proof. Assume $C^\mathcal{E} \subseteq D^\mathcal{E}$. Then $C^\mathcal{E} \cap R_r \subseteq D^\mathcal{E} \cap R_r$. Affine maps preserve inclusion, so $T_r(C^\mathcal{E} \cap R_r) \subseteq T_r(D^\mathcal{E} \cap R_r)$, and box addition with E_r preserves inclusion, giving $T_r(C^\mathcal{E} \cap R_r) \oplus E_r \subseteq T_r(D^\mathcal{E} \cap R_r) \oplus E_r$, i.e. $(\exists r.C)^\mathcal{E} \subseteq (\exists r.D)^\mathcal{E}$. $\square$

Proposition 3. For $r, s \in \mathbf{N}_R$, the following are equivalent: (1) $r^\mathcal{E} \subseteq s^\mathcal{E}$; (2) $R_r \subseteq R_s$ and $T_r(\mathbf{y}) \oplus E_r \subseteq T_s(\mathbf{y}) \oplus E_s$ for all $\mathbf{y} \in R_r$; (3) the same conditions for $\mathbf{y} \in \{\mathbf{l}_r^R, \mathbf{u}_r^R\}$.

Proof. Suppose $R_r \subseteq R_s$ and $T_r(\mathbf{y}) \oplus E_r \subseteq T_s(\mathbf{y}) \oplus E_s$ throughout R_r , and let $(\mathbf{x}, \mathbf{y}) \in r^\mathcal{E}$, so $\mathbf{y} \in R_r$ and $\mathbf{x} \in T_r(\mathbf{y}) \oplus E_r$. Then $\mathbf{y} \in R_s$ and $\mathbf{x} \in T_s(\mathbf{y}) \oplus E_s$, so $(\mathbf{x}, \mathbf{y}) \in s^\mathcal{E}$.

Conversely, suppose $r^\mathcal{E} \subseteq s^\mathcal{E}$ and let $\mathbf{y} \in R_r$. Since E_r is non-empty there is some $\mathbf{x}$ with $(\mathbf{x}, \mathbf{y}) \in r^\mathcal{E} \subseteq s^\mathcal{E}$, giving $\mathbf{y} \in R_s$; as $\mathbf{y}$ was arbitrary, $R_r \subseteq R_s$. Every $\mathbf{x} \in T_r(\mathbf{y}) \oplus E_r$ likewise lies in $T_s(\mathbf{y}) \oplus E_s$.

Finally, $T_r(\mathbf{y}) \oplus E_r \subseteq T_s(\mathbf{y}) \oplus E_s$ states componentwise that $\mathbf{m}_s \mathbf{y} + \mathbf{c}_s + \mathbf{l}_s^E \leq \mathbf{m}_r \mathbf{y} + \mathbf{c}_r + \mathbf{l}_r^E$ and $\mathbf{m}_r \mathbf{y} + \mathbf{c}_r + \mathbf{u}_r^E \leq \mathbf{m}_s \mathbf{y} + \mathbf{c}_s + \mathbf{u}_s^E$. Both are affine in $\mathbf{y}$ and separable across dimensions, so each holds throughout $[\mathbf{l}_r^R, \mathbf{u}_r^R]$ iff it holds at its two endpoints. $\square$

Proposition 4. If $r^\mathcal{E} \subseteq s^\mathcal{E}$, then role assertions transfer from r to s , and $(\exists r.C)^\mathcal{E} \subseteq (\exists s.C)^\mathcal{E}$ for every concept C .

Proof. The first claim is immediate from $r^\mathcal{E} \subseteq s^\mathcal{E}$. For the second, let $\mathbf{x} \in (\exists r.C)^\mathcal{E}$, so there is $\mathbf{y} \in C^\mathcal{E} \cap R_r$ with $\mathbf{x} \in T_r(\mathbf{y}) \oplus E_r$. By Proposition 3(2), $R_r \subseteq R_s$, so $\mathbf{y} \in C^\mathcal{E} \cap R_s$, and $T_r(\mathbf{y}) \oplus E_r \subseteq T_s(\mathbf{y}) \oplus E_s$, so $\mathbf{x} \in T_s(\mathbf{y}) \oplus E_s \subseteq (\exists s.C)^\mathcal{E}$. $\square$

Lemma 7. Let $\gamma \geq 0$ and let B_x, B_y be boxes. Then (1) $\ell_{\subseteq}^\gamma(B_x, B_y) = 0$ implies $B_x \subseteq B_y$; (2) if B_x, B_y are non-empty, $\ell_{\cap+}^\gamma(B_x, B_y) = 0$ implies $B_x \cap B_y \neq \emptyset$; and (3) if $\gamma > 0$, $\ell_{\cap-}^\gamma(B_x, B_y) = 0$ implies $B_x \cap B_y = \emptyset$. For non-empty B_x, B_y the converses of (1) and (2) hold at $\gamma = 0$.

Proof. Fix a dimension i and write $l = c - o$, $u = c + o$. Then $d_i + 2o_{x,i} = |c_{x,i} - c_{y,i}| + o_{x,i} - o_{y,i}$, so $d_i + 2o_{x,i} \leq 0$ iff $l_{y,i} \leq l_{x,i}$ and $u_{x,i} \leq u_{y,i}$, and $d_i \leq 0$ iff the projections of B_x and B_y onto dimension i meet.

Inclusion. The claim is trivial for $B_x = \emptyset$; otherwise zero loss gives $d_i + 2o_{x,i} \leq -\gamma \leq 0$ for all i , hence $B_x \subseteq B_y$. *Overlap.* Zero loss gives $d_i \leq -\gamma \leq 0$ for all i , hence $B_x \cap B_y \neq \emptyset$ for non-empty boxes. *Disjointness.* Zero loss gives $\max_i d_i \geq \gamma > 0$, so the projections are separated in some dimension j .

At $\gamma = 0$ the displayed conditions are exactly the vanishing conditions of the losses, giving both converses; the disjointness loss then also vanishes when the boxes meet at a boundary, which is why a strictly positive margin is used. $\square$

A.3. Role Composition

Fibres of the composite. Fix $\mathbf{z} \in R_{r_2}$ and let $F(\mathbf{z}) = \{\mathbf{x} \mid (\mathbf{x}, \mathbf{z}) \in r_1^\mathcal{E} \circ r_2^\mathcal{E}\}$. An intermediate point $\mathbf{y}$ must satisfy $\mathbf{y} \in T_{r_2}(\mathbf{z}) \oplus E_{r_2}$ and $\mathbf{y} \in R_{r_1}$, and then $\mathbf{x} \in T_{r_1}(\mathbf{y}) \oplus E_{r_1}$. Since T_{r_1} is diagonal affine with positive slope and box addition distributes over unions,

$$F(\mathbf{z}) = T_{r_1}((T_{r_2}(\mathbf{z}) \oplus E_{r_2}) \cap R_{r_1}) \oplus E_{r_1}. \quad (10)$$

Deleting the intersection gives the fibre of $\max(r_1 \circ r_2)$,

$$T_{r_1}(T_{r_2}(\mathbf{z}) \oplus E_{r_2}) \oplus E_{r_1} = (T_{r_1} \circ T_{r_2})(\mathbf{z}) \oplus \mathbf{m}_{r_1} E_{r_2} \oplus E_{r_1}. \quad (11)$$

This is of the form $\langle R, E, T \rangle$: $T_{r_1} \circ T_{r_2}$ is diagonal affine with slope $\mathbf{m}_{r_1} \mathbf{m}_{r_2} > \mathbf{0}$, and its error box is non-empty.

Proposition 5. With $\max(r_1 \circ r_2)$ and the compatibility condition as in Section 4.1: (1) $r_1^\mathcal{E} \circ r_2^\mathcal{E} \subseteq \max(r_1 \circ r_2)^\mathcal{E}$; (2) equality holds iff the compatibility condition holds; (3) if the compatibility condition fails, then $r_1^\mathcal{E} \circ r_2^\mathcal{E}$ is not of the form $\langle R, E, T \rangle$.

Proof. *Clause 1.* $(T_{r_2}(\mathbf{z}) \oplus E_{r_2}) \cap R_{r_1} \subseteq T_{r_2}(\mathbf{z}) \oplus E_{r_2}$, and T_{r_1} and box addition with E_{r_1} preserve inclusion, so (10) is contained in (11) for every $\mathbf{z} \in R_{r_2}$. Both relations have second argument ranging over R_{r_2} , hence $r_1^\mathcal{E} \circ r_2^\mathcal{E} \subseteq \max(r_1 \circ r_2)^\mathcal{E}$.

Clause 2. The two fibres over $\mathbf{z}$ coincide iff the intersection in (10) is the whole box, i.e. iff $T_{r_2}(\mathbf{z}) \oplus E_{r_2} \subseteq R_{r_1}$. Since $T_{r_2}(R_{r_2}) \oplus E_{r_2} = \bigcup_{\mathbf{z} \in R_{r_2}} (T_{r_2}(\mathbf{z}) \oplus E_{r_2})$, this holds for every $\mathbf{z} \in R_{r_2}$ iff the compatibility condition holds.

Clause 3. Every role $\langle R, E, T \rangle$ has non-empty fibres of constant width $\mathbf{u}^E - \mathbf{l}^E$. By (10) the composite's fibre width in dimension i is $\mathbf{m}_{r_1, i} w_i(z_i)$ up to a constant, where $w_i(z_i)$ is the width of $(T_{r_2}(\mathbf{z}) \oplus E_{r_2})_i \cap R_{r_1, i}$. Since $(T_{r_2}(\mathbf{z}) \oplus E_{r_2})_i$ has fixed width and translates at rate $\mathbf{m}_{r_2, i} > 0$, w_i is trapezoidal in z_i , so on an interval of positive width it is constant only where it vanishes or at its plateau. Where it vanishes the composite has an empty fibre, which already excludes the form $\langle R, E, T \rangle$. At the plateau either $(T_{r_2}(\mathbf{z}) \oplus E_{r_2})_i \subseteq R_{r_1, i}$, contradicting the failure of compatibility in dimension i , or the reverse inclusion holds throughout, making the fibre constant in z_i and hence a translate only for slope 0, excluded by $\mathbf{m} \in \mathbb{R}_{>0}^n$. $\square$

Proposition 6. If $\max(r_1 \circ r_2)^\mathcal{E} \subseteq s^\mathcal{E}$, then (1) $\mathcal{E} \models r_1 \circ r_2 \sqsubseteq s$; and (2) for concept descriptions X, C, D , if $X^\mathcal{E} \subseteq (\exists r_1.C)^\mathcal{E}$ and $C^\mathcal{E} \subseteq (\exists r_2.D)^\mathcal{E}$, then $X^\mathcal{E} \subseteq (\exists s.D)^\mathcal{E}$.

Proof. Assume $\max(r_1 \circ r_2)^\mathcal{E} \subseteq s^\mathcal{E}$.

Clause 1. By Proposition 5(1), $r_1^\mathcal{E} \circ r_2^\mathcal{E} \subseteq \max(r_1 \circ r_2)^\mathcal{E} \subseteq s^\mathcal{E}$, which is the satisfaction condition for $r_1 \circ r_2 \sqsubseteq s$.

Clause 2. Unfolding existential restriction twice gives, for any concept D ,

$$(\exists r_1.(\exists r_2.D))^\mathcal{E} = \{\mathbf{x} \mid \exists \mathbf{y} \in (\exists r_2.D)^\mathcal{E}, (\mathbf{x}, \mathbf{y}) \in r_1^\mathcal{E}\} = \{\mathbf{x} \mid \exists \mathbf{z} \in D^\mathcal{E}, (\mathbf{x}, \mathbf{z}) \in r_1^\mathcal{E} \circ r_2^\mathcal{E}\}.$$

From $C^\mathcal{E} \subseteq (\exists r_2.D)^\mathcal{E}$ and Proposition 2, $X^\mathcal{E} \subseteq (\exists r_1.C)^\mathcal{E} \subseteq (\exists r_1.(\exists r_2.D))^\mathcal{E}$. Each $\mathbf{x}$ in the latter has some $\mathbf{z} \in D^\mathcal{E}$ with $(\mathbf{x}, \mathbf{z}) \in r_1^\mathcal{E} \circ r_2^\mathcal{E} \subseteq s^\mathcal{E}$, hence $\mathbf{x} \in (\exists s.D)^\mathcal{E}$. Therefore $X^\mathcal{E} \subseteq (\exists s.D)^\mathcal{E}$. $\square$

An alternative approximation. Restricting instead to the region on which the intersection in (10) is vacuous gives the same map and error box as $\max(r_1 \circ r_2)$ but the eroded range

$$R_{r_1 \circ r_2} = R_{r_2} \cap T_{r_2}^{-1}(R_{r_1} \ominus E_{r_2}), \quad B_x \ominus B_y = [\mathbf{l}_x - \mathbf{l}_y, \mathbf{u}_x - \mathbf{u}_y], \quad (12)$$

an under-approximation of the composite, for which $R_\circ$ may fail. In one dimension take $B_C = [0.4, 0.6]$, $R_{r_1} = R_{r_2} = [0, 1]$, $T_{r_1} = T_{r_2} = \text{id}$, $E_{r_2} = [-0.6, 0.6]$ and $E_{r_1} = [-0.1, 0.1]$. Then $(\exists r_2.C)^\mathcal{E} = [-0.2, 1.2]$ and $(\exists r_1.(\exists r_2.C))^\mathcal{E} = [-0.1, 1.1]$, while $R_{r_1} \ominus E_{r_2} = [0.6, 0.4] = \emptyset$, so $R_{r_1 \circ r_2} = \emptyset$ and the restricted composite is empty. It is then contained in $s^\mathcal{E}$ for every role s , including those with $(\exists r_1.(\exists r_2.C))^\mathcal{E} \not\subseteq (\exists s.C)^\mathcal{E}$, so the conclusion of $R_\circ$ does not follow.

A.4. Soundness and Closure

Theorem 8. Let $\mathcal{O}$ be a normalised ontology and $\mathcal{E}$ an embedding in which every TBox axiom and every positive ABox assertion of $\mathcal{O}$ has zero geometric loss. Then $\mathcal{E}$ is a model of $\mathcal{O}$.

Proof. Since $\mathcal{O}$ is normalised it suffices to check each form of Table 2. Lemma 7 converts zero primitive losses into box conditions; where a left-hand box is empty the required inclusion holds trivially, so we may assume all boxes non-empty.

GCI0. $\ell_{\sqsubseteq}^\gamma(B_C, B_D) = 0$ gives $B_C \subseteq B_D$. *GCI0-BOT.* B_A is fixed to the canonical empty box, so $A^\mathcal{E} = \emptyset = \perp^\mathcal{E}$. *GCI1.* The inclusion term gives $B_{C_1} \cap B_{C_2} \subseteq B_D$, and $(C_1 \sqcap C_2)^\mathcal{E} = B_{C_1} \cap B_{C_2}$. *GCI1-BOT.* $\ell_{\sqcap}^\gamma(B_{C_1}, B_{C_2}) = 0$ gives $B_{C_1} \cap B_{C_2} = \emptyset = \perp^\mathcal{E}$. *GCI2.* The inclusion term gives $B_C \subseteq (\exists r.D)^\mathcal{E}$. *GCI3.* The inclusion term gives $(\exists r.C)^\mathcal{E} \subseteq B_D$. *GCI3-BOT.* $\ell_{\sqcap}^\gamma(R_r, B_C) = 0$ gives $B_C \cap R_r = \emptyset$, hence $(\exists r.C)^\mathcal{E} = \emptyset$. *RI0.* Zero loss gives $R_r \subseteq R_s$ and $T_r(\mathbf{y}) \oplus E_r \subseteq T_s(\mathbf{y}) \oplus E_s$ at the two endpoints of R_r , equivalent by Proposition 3 to $r^\mathcal{E} \subseteq s^\mathcal{E}$. *RI1.* The loss is the RI0 loss applied to $\max(r_1 \circ r_2)$ and s , so zero loss gives $\max(r_1 \circ r_2)^\mathcal{E} \subseteq s^\mathcal{E}$, and Proposition 6(1) gives $\mathcal{E} \models r_1 \circ r_2 \sqsubseteq s$. *Concept assertions.* $\ell_{\sqsubseteq}^\gamma([\mathbf{e}_a, \mathbf{e}_a], B_C) = 0$ gives $\mathbf{e}_a \in B_C$. *Role assertions.* Zero loss on the two inclusion terms gives $\mathbf{e}_b \in R_r$ and $\mathbf{e}_a \in T_r(\mathbf{e}_b) \oplus E_r$, which is $(a^\mathcal{E}, b^\mathcal{E}) \in r^\mathcal{E}$. $\square$

Corollary 9. Under the assumptions of Theorem 8, $\mathcal{E}$ satisfies every axiom derivable from $\mathcal{O}$ by the rules of Figure 1.

Proof. By Theorem 8, $\mathcal{E}$ is a model of $\mathcal{O}$; it remains to check each rule. R_0 and $R_\sqsubseteq$ are reflexivity and transitivity of box inclusion, and $R_\top$ holds since $C^\mathcal{E} \subseteq \mathbb{R}^n = \top^\mathcal{E}$. For $R_\perp$, if $C^\mathcal{E} = \emptyset$ then $(\exists r.C)^\mathcal{E} = T_r(\emptyset) \oplus E_r = \emptyset$, so $X^\mathcal{E} \subseteq (\exists r.C)^\mathcal{E}$ forces $X^\mathcal{E} = \emptyset$. $R_{\sqcap}^-$ and $R_{\sqcap}^+$ follow from $(D_1 \sqcap D_2)^\mathcal{E} = D_1^\mathcal{E} \cap D_2^\mathcal{E}$ and the characterisation of intersection as the greatest lower bound under inclusion. $R_\exists$ is Proposition 2, R_H is Proposition 4(2), and $R_\circ$ is Proposition 6(2). $\square$

A.5. Complexity

An embedding stores $2n|N_C| + n|N_I| + 6n|N_R|$ parameters ($2n$ per concept box, n per individual, $6n$ per role); the diagonal affine map keeps roles linear in n rather than $O(n^2)$. Scoring an axiom takes $O(n)$ box operations, so a TBox epoch is $O(k_{\text{neg}}n|\mathcal{T}|)$ for k_{neg} corruptions per axiom, and ranking a query against all candidates is $O(n|N_C|)$. The lexical term of Section 4.4 adds $O(m^2n)$ per batch of m individuals.

A.6. Benchmark Statistics

Table 6

Axiom counts by normal form in the normalised training ontologies, with class and role signature sizes.

	GCI0	GCI0-BOT	GCI1	GCI1-BOT	GCI2	GCI3	GCI3-BOT	RI0	RI1	$ N_C / N_R $
GALEN	28,890	0	13,595	0	28,118	13,597	0	958	58	23,141 / 950
GO	85,471	0	12,131	30	20,324	12,129	0	3	6	45,893 / 8
Anatomy	121,779	2	2,121	184	152,289	2,121	22	33	31	106,258 / 166

A.7. ESM-2 Representations

ESM-2 is a family of transformer-based protein language models trained on amino-acid sequences [23]. Let Σ be the amino-acid alphabet. For a sequence $p = (a_1, \dots, a_L)$ with $a_i \in \Sigma$, ESM-2 returns contextual residue embeddings $(\mathbf{h}_1, \dots, \mathbf{h}_L)$, each $\mathbf{h}_i \in \mathbb{R}^d$ computed with respect to the whole sequence. We mean-pool them to obtain a fixed-dimensional representation

$$\mathbf{z}_p = \frac{1}{L} \sum_{i=1}^L \mathbf{h}_i,$$

and use $\mathbf{z}_p$ as the external representation of p in Eq. (7). In our experiments $d = 480$.

A.8. Protein-Function Prediction Metrics

We use the protein-centric metrics of [31]. For a protein p , ELK-Em scores every candidate assertion $G(p)$ with G a GO concept; at a threshold t the predicted annotation set is $P_t(p) = \{G \in \mathcal{C} \mid s(G(p)) \geq t\}$, where $\mathcal{C}$ is the candidate set. Predictions and ground truth are both propagated through the GO hierarchy; $\Gamma(p)$ denotes the propagated ground-truth set. Then

$$\text{pr}_p(t) = \frac{|P_t(p) \cap \Gamma(p)|}{|P_t(p)|}, \quad \text{rc}_p(t) = \frac{|P_t(p) \cap \Gamma(p)|}{|\Gamma(p)|}.$$

Proteins with $P_t(p) = \emptyset$ are omitted from the precision average. Writing N for the number of evaluated proteins and $m(t)$ for the number with at least one prediction,

$$\text{Pr}(t) = \frac{1}{m(t)} \sum_{p: |P_t(p)| > 0} \text{pr}_p(t), \quad \text{Rc}(t) = \frac{1}{N} \sum_{p=1}^N \text{rc}_p(t), \quad F_1(t) = \frac{2 \text{Pr}(t) \text{Rc}(t)}{\text{Pr}(t) + \text{Rc}(t)}.$$

Thresholds are swept over 71 values evenly spaced in $[0.3, 1.0]$. We select the threshold maximising $F_1(t)$ on the validation split and apply it unchanged to the test split, reporting test precision, recall and F_1 at that operating point.

Table 7

Diagnostic variants on the human CAFA5 test split, single seed and therefore indicative rather than directly comparable to Table 4. The sweep varies the coefficient on $\mathcal{L}_{\text{lex}}$; the transfer rows apply nearest-neighbour annotation transfer, scoring by s^{geo} on ELK-Em’s trained individual embeddings and by cosine similarity on ESM-2 vectors.

Variant	Test F_1
Lexical weight 1	0.2466
Lexical weight 10	0.3036
Lexical weight 50	0.2668
Lexical weight 200	0.2466
Nearest-neighbour transfer, learned space	0.4327
Nearest-neighbour transfer, ESM-2 space	0.5932

A.9. Diagnostics for Inductive Prediction

These diagnostics support the analysis of Section 5.2: the weight on $\mathcal{L}_{\text{lex}}$ does not account for the gap to the baseline, and external structure transfers globally but not locally.

Spearman rank correlation between $\bar{s}^{\text{ext}}$ and s^{geo} over sampled pairs of individuals is 0.88 among training individuals and 0.96 between test and training individuals, against approximately zero at initialisation. The mean overlap between top- k_{nn} neighbourhoods in ESM-2 space and in the learned space ranges from 10% to 27% for k_{nn} between 1 and 50.